

5.4.6

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Question

Using elementary transformations, find the inverse of the following matrix.

$$\begin{pmatrix} 10 & -2 \\ -5 & 1 \end{pmatrix}$$

Solution

We employ the Gauss–Jordan method,

$$\left(\begin{array}{cc|cc} 10 & -2 & 1 & 0 \\ -5 & 1 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \leftarrow \frac{1}{10} R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{5} & \frac{1}{10} & 0 \\ -5 & 1 & 0 & 1 \end{array} \right) \quad (1)$$

$$\xrightarrow{R_2 \leftarrow R_2 + 5R_1} \left(\begin{array}{cc|cc} 1 & -\frac{1}{5} & \frac{1}{10} & 0 \\ 0 & 0 & \frac{1}{2} & 1 \end{array} \right) \quad (2)$$

At this stage, the left-hand side is not the identity matrix.
Hence the given matrix is not invertible.

Code (C)

Part 1

```
#include <stdio.h>

void compute_inverse(double* inv) {
    double a = 10, b = -2;
    double c = -5, d = 1;
    double det = a*d - b*c;

    if(det == 0) {
        inv[0] = inv[1] = inv[2] = inv[3] = 0;
    } else {
        inv[0] = d/det; // [0,0]
        inv[1] = -b/det; // [0,1]
        inv[2] = -c/det; // [1,0]
        inv[3] = a/det; // [1,1]
    }
}
```

Part 1

```
import ctypes
import numpy as np
import os

# Load the shared library
if os.name == "nt":
    lib = ctypes.CDLL("./code9.dll")
else:
    lib = ctypes.CDLL("./code9.so")

# Create a 2x2 NumPy array to hold the inverse
inv_matrix = np.zeros((2,2), dtype=np.double)

# Set the argument type for ctypes
lib.compute_inverse.argtypes = [ctypes.POINTER(ctypes.c_double)]
lib.compute_inverse.restype = None
```

Part 2

```
# Call the C function with pointer to NumPy array
lib.compute_inverse(inv_matrix.ctypes.data_as(ctypes.POINTER(
    ctypes.c_double)))

# Check if matrix is invertible
if np.all(inv_matrix == 0):
    print("Matrix is not invertible.")
else:
    print("Matrix is invertible.")
    print("Inverse matrix:\n", inv_matrix)
```